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A Machine Learning Approach to Optimize Drilling Operations in Real-Time

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Abstract

In the oil and gas drilling industry, optimizing key operational parameters—such as weight on bit (WOB), rate of penetration (ROP), and Revolutions Per Minute (RPM)—is essential for maximizing drilling efficiency, minimizing costs, and ensuring operational safety. Drilling environments are highly dynamic, influenced by constantly changing geological formations, mud properties, and downhole conditions. This research addresses the critical need for real-time optimization of drilling parameters. Despite significant advancements in drilling automation, many existing systems still lack the adaptability to respond effectively to changing downhole conditions and transparency in decision-making. This study proposes an adaptive machine learning (ML) and optimization model framework to bridge this gap. The goal is to both validate the performance of models in dynamic drilling environments and assess how interpretability methods can foster operator trust. Using the dataset provided, the framework simulates real-time adaptability through an offline, condition-segmented analysis approach. The dataset is divided into subsets representing distinct drilling conditions, such as variations in depth or Mud Weight (MW). For each condition, regression and time series models are trained to predict the rate of penetration, and an optimization algorithm is then implemented to continuously provide optimum controllable parameters, for maximum penetration rates within safe Formation Pressure ranges. The results show significant improvements in drilling efficiency and cost reduction, demonstrating the potential of the proposed framework for real-time optimization. Additionally, the study highlights the importance of integrating ML and optimization techniques in drilling operations, enabling more accurate and informed decision-making, and paving the way for further advancements in autonomous drilling systems and digital transformation in the oil and gas industry, with potential applications in various drilling scenarios.

Keywords: Drilling, Prediction, ROP, Optimization, Machine Learning

Introduction

Drilling operations lie at the core of oil and gas exploration, yet they remain among the most cost-intensive and technically demanding stages of well construction. The complexity of drilling operations lies not

only from the challenge of overcoming difficult geological conditions but also from the need to maintain operational safety, efficiency, and regulatory compliance in a dynamic and unpredictable environment. Even minor inefficiencies in setting and adjusting key operational parameters—such as Weight on Bit (WOB), Rate of Penetration (ROP), and Revolutions Per Minute (RPM)—can quickly accumulate, leading to longer drilling times, higher equipment wear, and increased exposure to safety and environmental risks. These challenges are further amplified in complex wells, where operational parameters must continuously adapt to changing geological formations, fluctuating mud properties, and evolving downhole conditions. At the same time, the oil and gas industry face increasing demands to improve efficiency, enhance safety, and operate sustainably in drilling operations. Traditional drilling practices, heavily reliant on manual decision-making and static, pre-programmed models, often struggle to adapt to real-time variations in subsurface conditions. This rigidity contributes to operational inefficiencies, elevated costs, and heightened risks of operational failure or well integrity issues.

In recent years, drilling automation systems have been introduced to improve performance by leveraging surface data and pre-defined algorithms. While these systems have yielded measurable gains, they often lack the adaptability required to keep pace with rapidly shifting downhole realities. Models trained on historical datasets tend to lose accuracy when confronted with unexpected lithological changes, and their opaque "black box" decision-making limits operator trust and adoption. Recent advancements in machine learning (ML) offer a transformative opportunity to overcome these barriers. By processing high-frequency data from drilling sensors, and historical records, ML models can dynamically predict critical parameters, and recommend optimal operational adjustments in real time.

This paper builds on these developments by proposing an adaptive ML framework aimed at optimizing four controllable inputs WOB, flow rate (FR), mud weight (MW), and RPM, with the goal of maximizing the RPM, which in return improves drilling efficiency, and reduces costs while remaining in the safe range of operations.

Overview on the paper structure

The next sections of this paper are organized as follows: The next section reviews recent literature on drilling optimization, automation systems, and the emerging role of ML in well operations. Following that, the methodology section outlines the proposed adaptive ML framework, detailing the data sources, and real-time learning approach. After that, the section discusses the results, highlighting performance improvements, operational implications, and key lessons learned. Finally, the last section concludes the paper by summarizing the main findings and outlining opportunities for future research.

Problem Definition and Objectives

Problem Statement

Drilling operations are highly sensitive to unpredictable subsurface conditions caused by variations in rock formations, pore pressure, and fluid content. Such uncertainty can lead to inefficient drilling paths, accelerated tool wear, or even serious hazards such as blowouts. To mitigate these risks, constant adjustment of key parameters, including WOB, ROP, RPM, FR, and MW is essential. However, current control practices whether manual or based on static models struggle to adapt quickly to real-time changes in downhole conditions. This lack of responsiveness results in inefficiencies, higher operational costs, equipment damage, and elevated safety risks. While existing drilling automation systems have improved performance, they remain constrained by historical models that degrade under unexpected geological changes and by their opaque decision-making processes, which limit operator trust and widespread adoption.

Objectives

This research directly addresses these challenges by implementing a ML model specifically designed to optimize drilling operations in real time. The framework focuses on maximizing ROP while dynamically adjusting four controllable inputs: WOB, FR, MW, and RPM.

The objectives are to:

- Optimize drilling parameters in real time to achieve the highest possible ROP without compromising well integrity
- Maintain safety boundaries by ensuring all optimized parameter values remain within the safe fracture pressure range to prevent blowouts or other hazards
- Continuously respond to downhole changes by allowing the ML model to update inputs dynamically as new data is received, ensuring drilling operations remain efficient and safe under varying geological conditions

By combining adaptive optimization with built-in safety constraints, the proposed ML model provides a data-driven approach to achieving faster, safer, and more efficient drilling operations.

Literature Review

Drilling optimization is a crucial element in ensuring the sustainability of oil and gas operations, as it enhances efficiency, reduces costs, and promotes safer working conditions. Drilling itself is among the most critical and expensive phases of field development, often consuming over half the project budget. Rising costs have driven engineers to focus on reducing drilling time while maximizing performance. Efforts typically target controllable parameters (WOB, RPM, and FR) as these directly influence the ROP. However, simply maximizing ROP is not always beneficial; excessive drilling speed can lead to poor hole cleaning, damage drill bits, and create wellbore instability. Over time, engineers have developed different strategies to optimize these parameters starting with manual tuning and empirical models, then moving to statistical frameworks, automated systems, and ultimately adaptive ML approaches. The following sections trace this evolution, starting with early parameter-based optimization methods.

Early Approaches to Drilling Optimization

Early drilling optimization efforts relied heavily on deterministic and statistical models rather than ML. These approaches focused on predicting ROP and adjusting key parameters such as WOB, RPM, and FR, which had strong results, but important limitations.

AlHinaai [1] developed a real-time optimization system, using regression modeling to dynamically control parameters like hole depth, bit weight, FR, pump pressure, block height, top drive RPM, and torque. The resulting model achieved an impressive prediction accuracy of over 90% when compared to actual field measurements. Additionally, the model was enhanced with an advanced bit tooth wear model to optimize both ROP and the operational lifespan of the drill bit. Despite being real-time, it remained regression-based and limited in adaptability. Similarly, Bani-Mustafa et al. [2] used response surface methodology (RSM) to statistically model how WOB, RPM, and FR influenced ROP. Their findings confirmed that FR had the strongest effect, while excessive WOB could reduce efficiency. But RSM is inherently static, meaning it can identify optimal ranges from historical data but cannot update itself when downhole conditions change.

Oedegaard et al. [3] pushed further by designing an edge-based system combining cloud computing and AI-driven modeling to recommend parameter adjustments. The system featured auto-calibration, updating its predictions as conditions shifted, but was still largely advisory, without a clear optimization goal like maximizing ROP or minimizing energy use.

Alongside these efforts, Sobhi et al. [4] demonstrated how classical ROP equations could be enhanced using metaheuristic optimization. Testing models like Maurer, Bingham, and Bourgoyne & Young (B&Y)

against field data from Louisiana and Algeria, they found B&Y most accurate. By applying calibration techniques from multiple regression and Excel's GRG solver to Ant Colony Optimization (ACO) and MATLAB tools they achieved R^2 values up to 0.98 and root mean square error (RMSE) as low as ~ 2.85 , proving that even traditional models could yield robust ROP forecasts when paired with advanced tuning. Al-Rubaii et al. [5] built on this foundation by integrating bit hydraulics and mud rheology into a new ROP model. Their three-phase methodology revealed that higher hydraulic horsepower and jet velocity improved cutting transport and ROP by up to 50%. Yet, like other early models, it remained offline and static, requiring recalibration and offering no real-time self-correction.

Taken together, these studies show how engineers moved from manual and empirical tuning to more advanced statistical and advisory systems. They delivered meaningful improvements but remained static, semi-manual, and constrained by historical data, before the shift to automation, IoT frameworks, and ultimately, ML driven systems.

Advancements in Drilling Operations and Parameter Optimization

The next wave in drilling optimization came with automation and digitalization, gradually layering intelligence over traditional methods.

Teodoriu and Bello [6] provide a comprehensive review of how the industry evolved from manual practices to semi-autonomous systems. Their article traces how incremental technological milestones—such as wired drill pipe telemetry, real-time downhole sensors, and platforms which provided early advisory systems and set the stage for fully automated workflows. These systems began by offering drillers recommendations on WOB, RPM, and FR, and later evolved into feedback loops that could autonomously adjust parameters based on incoming sensor data. Their review also highlights digital twin frameworks, which mirror the wellbore in real time, combining physics-based models with live sensor data to predict hazards and optimize drilling ahead of the bit. While automation brought efficiency gains, Teodoriu & Bello emphasize enduring gaps: many systems remain semi-manual, lack embedded safety envelopes, and offer limited transparency in their decision logic, setting the stage for machine learning-driven systems to address these weaknesses.

Adding to this progression, the IEEE paper by Gooneratneet et al. [7] presents an in-depth vision for how Internet of Things (IoT) technologies and digital transformation (DT) can reshape drilling operations by moving from siloed, manual processes toward integrated, intelligent systems. The authors argue that decades of reliance on legacy tools and conservative practices have slowed adoption of disruptive technologies in oil and gas, but rising economic pressures and technological advancements now make automation inevitable. Their methodology focuses on combining real-time sensor data—from surface hydraulic sensors, downhole measurement-while-drilling (MWD) and logging-while-drilling (LWD) tools—into an IoT framework where data can be processed, analyzed, and acted upon locally through edge/fog computing, edge computing processes data locally on rigs, mitigating cybersecurity risks and enabling high-frequency analytics critical for real-time drilling control. This reduces dependence on slow, expensive satellite links and enables high-frequency data streams (e.g., WOB, RPM, gamma ray, mud rheology) to feed analytics and AI models directly on the rig.

The paper highlights a vision for gradual digital mastery, moving from "fashionista" adoption of scattered tools to integrated ecosystems combining IoT, AI/ML, and advanced analytics. Objectives include cutting upstream operating expenditures by 20–30% and improving drilling efficiency by at least 5%, while findings show that smart sensors, connected infrastructure, and automated logistics could revolutionize rig performance.

Building on this, Ochulor et al. [8] compare how new technologies are reshaping both onshore and offshore drilling. Onshore operations increasingly rely on automation, robotics, and real-time data systems that allow engineers to adjust drilling parameters instantly, simulate well paths through digital twins, and use predictive models to avoid downtime. Case studies from shale fields, such as the Permian Basin,

show how these tools improve efficiency, cut costs, and strengthen wellbore stability. Offshore drilling faces different challenges: deepwater, remote access, and harsh environments, so the focus shifts to subsea systems, advanced rigs with dynamic positioning, and remote monitoring centers. Remotely operated vehicles (ROVs) and satellite-linked controls are used to manage risks and reduce the need for personnel on site. Looking ahead, the paper highlights AI, machine learning, advanced materials, and renewable energy integration as promising directions. Overall, the study highlights how tailored innovations address the unique technical and environmental demands of onshore versus offshore drilling.

Sharma et al. [9] deliver a comprehensive review of how the Industrial Internet of Things (IIoT) is transforming operations across the oil and gas sector, from upstream exploration to downstream distribution. The paper highlights critical IIoT applications such as real-time monitoring, predictive maintenance, enhanced health, safety, and environmental (HSE) compliance, inventory control, and pipeline security, including the detection of oil theft and gas leaks. It carefully addresses deployment challenges, including concerns around data reliability, cyber-security vulnerabilities, and the integration of legacy systems. The authors conclude with practical recommendations: a clearly defined IIoT strategy, strong collaboration between IT and operational domains, robust data governance, cybersecurity investments, and ongoing optimization. Additionally, they emphasize future optimization through the integration of AI, machine learning, edge computing, augmented reality, and autonomous underwater systems to enhance resilience, safety, and environmental performance.

These studies illustrate how automation and IoT-driven digitalization paved the way for ML frameworks that can both adapt in real time and embed safety and explainability into drilling decisions.

Machine Learning Applications in Drilling Operations

AI-powered analytics are revolutionizing decision-making, streamlining drilling processes, and driving efficiency throughout the entire value chain from exploration and construction to production and maintenance. AI and ML enable the analysis of vast and complex datasets generated from drilling operations and equipment performance in real time. Actionable insights are extracted from various data sources, leading to efficient predictions for operational strategies across upstream activities. Here are some studies where researchers discussed the application of ML in drilling operations.

Alsaihati et al. [10] tackle one of the most significant challenges in drilling operations: downhole vibrations, which can severely impact drilling efficiency. To address this issue, the researchers proposed a framework that combines ML and optimization techniques to improve the ROP while simultaneously mitigating lateral and torsional vibrations. The proposed framework does not rely on downhole sensors, as their high cost prevents widespread use in every well. The ML component of the framework utilizes a Multiclass-Multioutput Classification (MMC) model to predict the severity of vibrations, along with a regression model to estimate ROP. Several ML algorithms were trained on historical well dataset and evaluated during the study, including K-Nearest Neighbors (K-NN), Decision Trees, Random Forest (RF), Gradient Boosting, and XGBoost. The evaluation results indicate that the RF model demonstrated the best overall performance. This model was integrated with the Differential Evolution (DE) optimization framework to refine drilling operations, and four optimization scenarios were executed to identify the optimal drilling configuration. The MMC model achieved a Jaccard similarity score of 0.83, reflecting a high level of consistency between the predicted and actual vibration classifications. Meanwhile, the regression model attained an R^2 score of 0.86 and a relatively low RMSE of 0.37. The proposed ML-optimization framework provides a robust and sensor-independent solution that not only improves operational performance but also significantly reduces costs by eliminating the need for expensive downhole sensors. This makes it a scalable and economically viable approach for future drilling operations.

Another paper by Olukoga and Feng [11] further advances the application of ML in drilling operations, aiming to enhance both efficiency and profitability. Unlike previous reviews that broadly examined ML applications in the oil and gas industry, their work specifically focuses on identifying drilling-related

challenges by analyzing recent studies and examining how ML techniques have been applied to address them. According to the case studies reviewed, artificial neural networks (ANN), support vector machines (SVM), and regression models are the most commonly applied ML algorithms in drilling applications. These models are primarily employed to predict critical drilling events such as ROP, Differential Pipe Sticking (DPS), drill string vibrations, and bit wear—enabling proactive decision-making and enhanced drilling performance.

The two studies highlight the growing role of ML in revolutionizing drilling operations and tackling long-standing challenges within the oil and gas sector. By leveraging advanced ML techniques, both research efforts demonstrate how data-driven solutions can enhance operational performance, increasing the profitability, and reducing costs of drilling operations.

Other Recent Challenges Related to Drilling Operations

Drilling operations in the oil and gas industry are highly complex and inherently risky. Recent challenges related to drilling operations have made it even more critical to follow robust safety protocols and best practices. These measures are essential not only to prevent accidents, injuries, and environmental harm but also to ensure compliance and support the long-term sustainability of drilling activities. Below are some studies that discussed the recent challenges related to drilling operations.

David et al. [12] published a study assessing the effectiveness of safety protocols and risk management strategies in minimizing operational hazards, with an emphasis on ensuring safe and sustainable practices across the entire drilling lifecycle, from exploration to production and decommissioning. The study proposed a comprehensive safety framework that highlights several key components. At its core is the Safety Management System (SMS), which serves as the foundational layer and includes policies and procedures to ensure all drilling activities align with established safety standards. The framework also integrates risk identification processes, leveraging predictive models to analyze historical data and anticipate potential incidents such as blowouts, equipment failures, and toxic gas releases. Emergency response and safety training are emphasized as essential elements of the framework. Furthermore, the study highlights the importance of fostering a strong safety culture across all levels of operations. Lastly, the integration of digital technologies such as real-time monitoring through IoT sensors, predictive maintenance, and automated safety systems is emphasized as a key enabler for enhancing operational safety and efficiency. The study outlines key challenges to consider when implementing the safety framework. These include the need for continuous improvement in safety protocols, which are influenced by technological advancements. Human behavior and insufficient training, regulatory compliance, and environmental considerations are also identified as critical factors in building sustainable safety practices.

In another study, Narayana [13] investigated the potential of integrating edge computing with cloud infrastructure to significantly enhance operational efficiency, safety, and decision-making in the oil and gas industry. Edge computing plays a key role by enabling real-time data processing by placing sensors, smart meters, and drones at remote field sites. By processing time-sensitive data locally, edge computing effectively reduces latency and bandwidth constraints, enabling faster and more strategic decision-making. In addition, the cloud integration provides a centralized infrastructure and computational power for applying advanced analytics and machine learning to large amount of data to predict equipment failures and enhance decision making. The paper discussed some of the main challenges related to Edge-Cloud synergy in the oil and gas industry. These include limited network reliability and connectivity, which makes it difficult to transmit data between edge devices and the cloud. Security and compliance are also concerns, as sensitive data could be exposed to cybersecurity threats. Another challenge is integrating these technologies with legacy systems, which may encounter various configuration issues.

Methodology

This study introduces a constrained optimization model designed to enhance drilling efficiency by maximizing the ROP while maintaining wellbore stability through safe formation pressure (FP) limits. The methodology integrates two Artificial Intelligence (AI) models—one for predicting ROP and the other for estimating FP—which are trained using both controllable drilling parameters (e.g., RPM, MW, FR, and WOB) and condition-based variables derived from open-source drilling datasets. These AI-driven predictive models capture the complex, nonlinear relationships between drilling inputs and operational responses. The optimization framework then utilizes these models to dynamically determine the optimal set of controllable parameters that maximize ROP while ensuring the drilling process remains within the safe operational envelope defined by FP constraints and ongoing downhole conditions.

The methodology consists of five key stages: Data Acquisition and Preparation, Development of a ROP and FP Prediction Model, Formulation of MW Constraint, Construction of the Optimization Model, and Validation and Sensitivity Analysis.

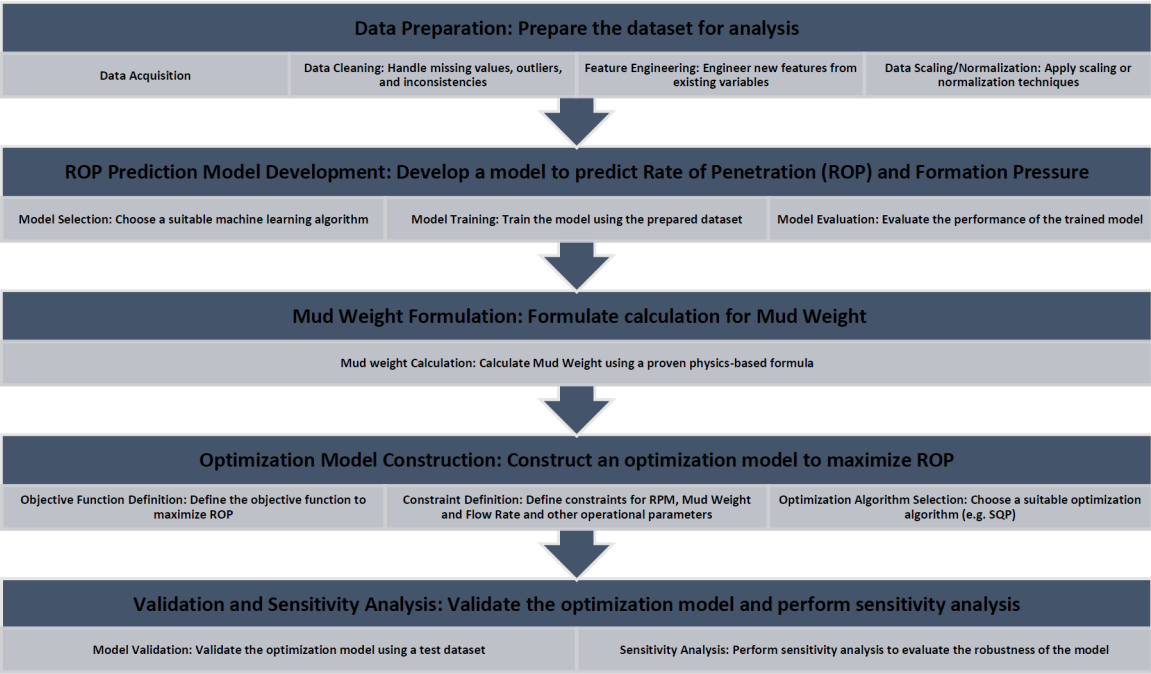


Figure 1—Overall Architecture

Data Acquisition and Preparation

The study utilizes an open-source dataset made available by Arifeen et al. (2024) [14], which contains drilling parameters and corresponding well performance data. The dataset is curated for tasks such as FP prediction and kick detection, and consists of 28 drilling-related variables with over 2000 data samples in each. The dataset encompasses a range of operational variables, including:

- Controllable parameters: RPM, WOB, and FR.
- Measured/Derived parameters: ROP, Formation Density (FDensity), Mud Viscosity (MVis), Standpipe Pressure (STP), Well Depth, and others
- Geological parameters: Relevant geological formations or properties, such as lithology and compressive strength

To prepare the dataset for analysis, the following steps were taken:

- **Data Cleaning:** The initial dataset was subjected to a rigorous cleaning process to handle missing values, outliers, and inconsistencies.
- **Feature Engineering:** New features were engineered from existing variables to improve the accuracy of the ROP and FP prediction models. The MW calculation is based on a proven physics-based formula commonly used in drilling operations.
- **Data Scaling/Normalization:** To ensure optimal model performance and prevent biases due to differing scales of input features, appropriate scaling or normalization techniques were applied.

These data preparation steps ensured that the dataset was accurate, complete, and consistent, providing a solid foundation for the development of the optimization model

Development of ROP and FP Prediction Models

The accurate prediction of ROP is a crucial factor in optimizing drilling efficiency. Additionally, the reliable estimation of FP is essential for ensuring safe drilling operations and preventing hazardous events such as kicks or blowouts. To achieve this, a reliable and robust prediction model is essential. In this section, we describe the development of both ROP and FP prediction models utilizing advanced machine learning techniques.

The goal of the model selection process was to identify the most accurate predictor of ROP. After careful consideration, a machine learning approach was adopted to develop predictive models for forecasting the ROP and FP. A supervised learning model was selected as the core architecture due to its strong predictive capability and adaptability to complex, data-driven relationships. Compared to traditional statistical models or simpler regression techniques, well-constructed ML models offer significant advantages—they can capture intricate nonlinear patterns, handle high-dimensional feature spaces, and generalize effectively across diverse operational conditions. Furthermore, advanced machine learning models provide improved accuracy through their ability to learn from historical data without relying on rigid assumptions about the underlying data distribution. When properly trained and validated, these models deliver reliable predictions, support better decision-making, and can be adapted to evolving drilling environments.

Once the model was selected, the training process commenced. The model was trained using 80% of the dataset, which provided a sufficient amount of data to learn the underlying patterns and relationships. The training process involved two critical steps: hyperparameter tuning and model evaluation. Hyperparameter tuning was performed using a grid search, which allowed for the optimization of the model's performance using a validation dataset. This ensured that the model was able to generalize well to unseen data and prevented overfitting.

The trained model's predictive accuracy was evaluated using the Mean Squared Error (MSE) on a separate test dataset (20% of the total dataset). Additionally, other performance metrics such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) were also used to assess the model's performance. These metrics provided a comprehensive understanding of the model's ability to accurately predict the targets.

In conclusion, the development of both prediction models utilizing the best models has resulted in a robust and reliable predictor. The model's ability to capture complex, nonlinear relationships within multivariate datasets and its accuracy in predicting the targets, makes it an essential tool for optimizing drilling operations. The use of advanced machine learning techniques and careful model selection and training have ensured that the model is well-suited for real-world applications, providing a significant improvement in drilling efficiency and safety.

Formulation of Mud Weight (MW)

MW plays a critical role in optimizing the ROP during drilling operations. It directly affects wellbore stability, FP control, and hydraulic efficiency.

The calculation of MW is based on the following equation:

$$MudWeight(MW) = (WBoPress - CPress) / (0.052 \times TVD) + BaseMudWeight$$

Where:

WBoPress = Well Bore Pressure (psi)

CPress = Casing Pressure (psi)

TVD = True Vertical Depth (which can be calculated using BTBR, Bit TVD below RKB) (ft)

Base MW = The density of the drilling mud (which can be calculated using AMTD, Active Mud Tank Density) (ppg)

Note: The 0.052 factor is a conversion factor to convert pressure to MW.

Using the provided fields, the MW calculation can be performed as follows [15] :

Calculate the True Vertical Depth (TVD):

TVD = BTBR (Bit TVD below RKB)

Calculate the Base MW:

Base MW = AMTD (Active Mud Tank Density)

Calculate the MW:

$MW = (WBoPress - CPress) / (0.052 \times TVD) + Base\ MW$

Proper MW prevents formation fluids from entering the wellbore, reducing the risk of kicks or blowouts. Additionally, it supports efficient hole cleaning by maintaining annular velocity and carrying cuttings to the surface. An optimized MW minimizes downhole complications, reduces bit wear, and enhances drilling efficiency. Therefore, incorporating MW into ROP prediction and optimization models is essential for safe, cost-effective, and high-performance drilling operations.

Construction of the Optimization Model

In this study, we employed a data-driven optimization approach to maximize the penetration rate of drilling operations. The optimization model was designed to identify the optimal combination of controllable parameters, including WOB, RPM, FR, and MW, to achieve the highest penetration rate while ensuring safe drilling operations.

Optimization Problem Formulation. The optimization problem was formulated as a minimization problem, where the objective function was defined as the negative of the ROP. The ROP was predicted using a trained machine learning model, which took into account various drilling parameters, including WOB, RPM, FR, MW, and other relevant variables. The optimization problem was subject to bounds on the controllable parameters, which were defined based on operational constraints and safety considerations.

Optimization Algorithm. The optimization problem was solved using the Differential Evolution (DE) algorithm, a global optimization technique that is well-suited for non-linear and non-convex problems. The DE algorithm was chosen for its ability to handle multiple local optima and its robustness to noise and uncertainty in the objective function. The algorithm was implemented with a seed value of 42 to ensure reproducibility of the results.

Optimization Model. The optimization model consisted of the following components:

Objective Function: The objective function was defined as the negative of the predicted ROP, which was calculated using the trained machine learning model.

Controllable Parameters: The controllable parameters included WOB, RPM, FR, and MW, which were optimized to maximize the ROP.

Bounds: The bounds on the controllable parameters were defined as follows:

WOB: [3463, 18936] pounds (lb)

RPM: [-0.057125, 0.000000] feet per minute (ft/min)

FR: [-0.615726, 0.000000] gallons per hour (gph)

MW: [0.037711, 0.061123] ppg (pounds per gallon)

Predicted ROP: The predicted ROP was calculated using the trained machine learning model, which took into account the optimized controllable parameters and other relevant drilling parameters.

Predicted FP: The predicted FP was calculated using a separate machine learning model, which took into account the optimized controllable parameters and other relevant drilling parameters.

Optimization Results. The optimization model was applied to a dataset of drilling operations, and the results were evaluated in terms of the predicted ROP and FP. The optimization results showed that the model was able to identify the optimal combination of controllable parameters that maximized the penetration rate while ensuring safe drilling operations. The results also highlighted the importance of considering multiple drilling parameters and their interactions in optimizing drilling operations.

Safe Drilling Operations. To ensure safe drilling operations, the optimization model included a constraint that checked whether the predicted FP was within the safe window defined by the pore pressure and fracture pressure. The results showed that the optimized drilling parameters resulted in a predicted FP that was within the safe window, indicating that the drilling operations were safe and did not pose a risk of wellbore instability or fracture.

Validation and Sensitivity Analysis

The validation of the optimization model is a crucial step in ensuring the accuracy and reliability of the proposed methodology. In this section, we describe how the optimized drilling parameters were validated through comparison with actual field data, and we evaluate the overall performance of the predictive model used in the optimization process.

To assess the effectiveness of the model, we compared the predicted ROP values generated by the optimization framework with the actual ROP values obtained from the testing validation dataset. This comparison provided a clear measure of how well the model generalizes to real-world drilling conditions and whether the optimized parameters yield performance improvements over baseline or conventionally used settings.

Model performance was evaluated using standard predictive metrics such as MAE and RMSE. These metrics allowed us to quantify the accuracy of the model's predictions and to benchmark its performance against acceptable industry standards. Low values of MAE and RMSE indicated that the model was capable of reliably predicting ROP under a range of drilling conditions.

In addition to numerical metrics, visual comparisons such as cross-plots of predicted versus actual ROP values were used to further validate the model's performance. These plots illustrated the degree of alignment between the model's predictions and the observed data, highlighting areas of strong agreement as well as any systematic deviations.

Furthermore, the optimized drilling parameters, such as WOB, RPM, and FR, were compared against the actual values used in the field. This comparison revealed opportunities for performance improvement and confirmed that the optimized parameters were both technically feasible and operationally relevant. In several cases, the optimized parameters led to a significant increase in ROP without compromising drilling safety or equipment integrity.

By systematically validating the model and comparing optimized results with actual field performance, we ensured that the proposed optimization methodology is not only theoretically sound but also practically effective in real drilling scenarios. This validation process reinforces the model's applicability for decision support and continuous improvement in drilling operations.

Result and Findings

The results of this study are presented in two main parts: the performance of the ROP prediction model and the outcomes of the drilling parameter optimization.

ROP Prediction model Performance

Table 1—Result Evaluation for the Model

Model	Best MSE	Best MAE	Best RMSE
ML 1	84.44554	8.939173	9.189425
ML 2	0.035245	0.037162	0.187737
ML 3	1147.773	33.80882	33.8788
ML 4	0.010243	0.034446	0.10121

In the evaluation of the predictive model for ROP, ML 4 emerged as the top-performing model with the lowest MSE of 0.010243 and RMSE of 0.10121, indicating highly accurate predictions with minimal error. This was closely followed by ML 2, which also demonstrated strong performance, achieving an MSE of 0.035245 and an RMSE of 0.187737. In contrast, ML 1 and ML 3 exhibited significantly higher error metrics, suggesting less reliable predictive capabilities. While ML 4 also recorded the lowest MAE of 0.034446, all models showed relatively small mean absolute errors, highlighting their general tendency to produce predictions close to the actual values. Overall, these findings underscore ML 4's superior effectiveness in capturing the underlying patterns within the dataset.

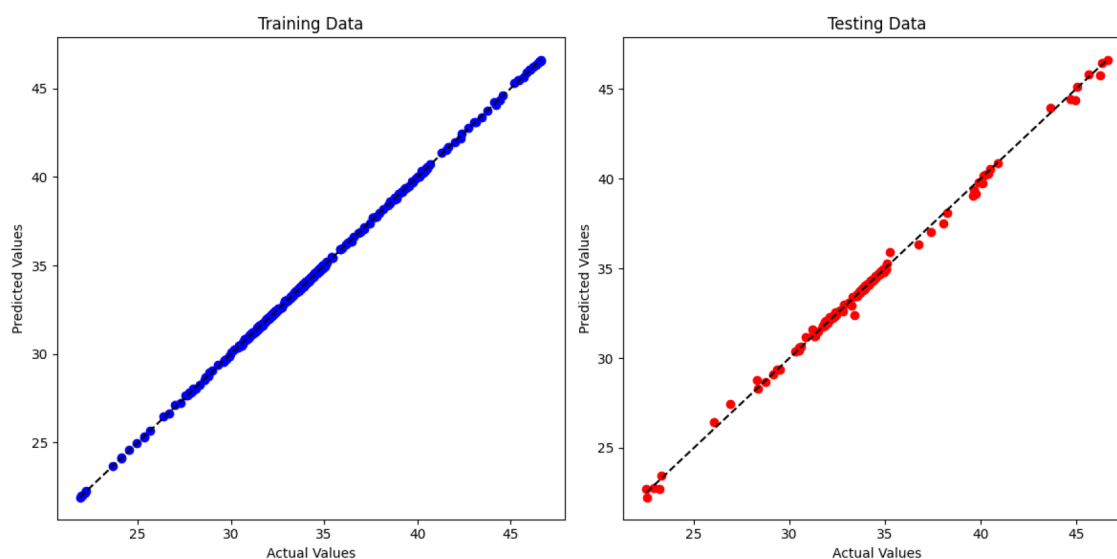


Figure 1a—ROP Prediction ML4 Comparison Training and Test Performance

The ML4 model shows excellent agreement between predicted and actual values for both training and testing datasets, with minimal scatter. Slightly higher variance in testing results is within acceptable limits, indicating strong generalization and no significant overfitting, making the model suitable for deployment.

Formation Pressure Prediction Model Performance

Table 2—Result Evaluation for the Model

Model	Best MSE	Best MAE	Best RMSE
ML 1	2361.46	15.59526	48.59489
ML 2	5297.59	11.11545	72.78457
ML 3	224445400.00	14945.8	14981.5

In the assessment of FP prediction models, the performance of the evaluated ML models reveals notable differences in predictive accuracy. ML 1 demonstrated the lowest MSE at 2361.46 and RMSE at 48.59, outperforming ML 2, which recorded higher MSE and RMSE values of 5297.59 and 72.78, respectively. In contrast, ML 3 exhibited significantly larger errors, with a Best MSE of 224,445,400 and Best RMSE of 14,981.5, indicating poor generalization and potential overfitting or instability in training. While ML 1 also showed a relatively low Mean Absolute Error (MAE) of 15.59, ML 2 achieved a slightly better MAE of 11.11, suggesting it may produce more balanced predictions in certain scenarios. Overall, ML 1 demonstrates superior performance for FP prediction under the given conditions, offering a more reliable basis for real-time pore pressure estimation and drilling decision support.

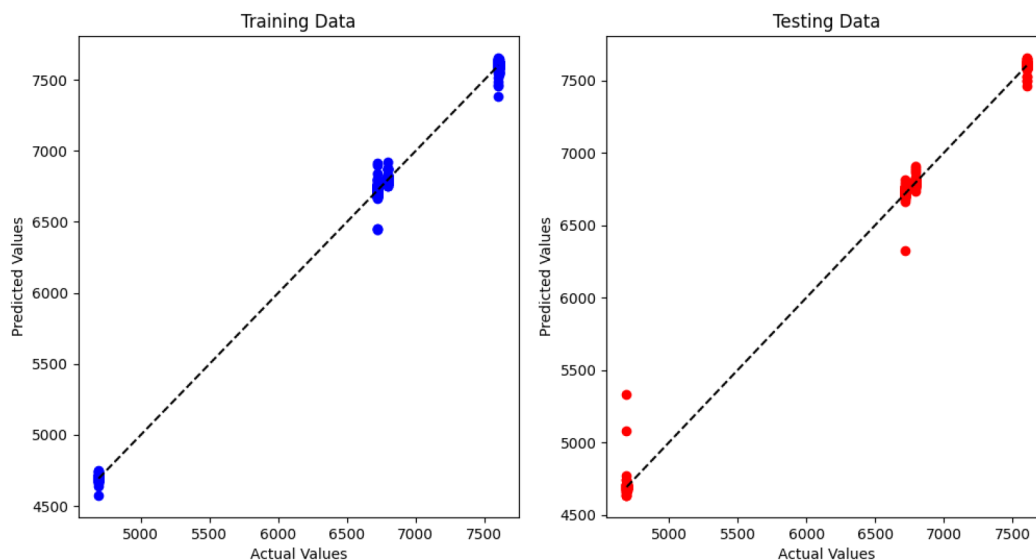


Figure 2—FP Prediction ML1 Comparison Training and Test Performance

The ML1 model demonstrates strong alignment between predicted and actual FPs in both training and testing datasets. Data points cluster tightly along the 45-degree line, with slightly more scatter in the test set. The performance gap is minimal, indicating robust generalization and no significant overfitting.

Drilling Parameter Optimization Outcomes. Building upon the predictive accuracy models, the optimization phase aimed to identify the optimal set of drilling parameters—namely, WOB, RPM, and FR—that would maximize the predicted ROP while adhering to operational constraints.

The optimization model demonstrated a measurable improvement in ROP compared to actual field performance. Figure 3 illustrates the frequency distribution of both actual and optimized ROP values. The actual ROP values (blue) are righted around a mean of 33.84, whereas the optimized ROP values (orange) are righted around a higher mean of 35.52.



Figure 3—Actual vs Optimized ROP

The shift in the distribution indicates that the optimization process consistently produced higher ROP values, with a mean improvement of 5.23% across all evaluated drilling intervals. This improvement is achieved without introducing significant variability, as the optimized ROP distribution remains compact and comparable in spread to the actual ROP distribution.

The histogram further reveals that the majority of optimized ROP values cluster between 34 and 36, while the actual ROP values are more concentrated between 33 and 35. This tighter upward shift suggests that the optimization model effectively identifies operational parameters that enhance drilling efficiency while maintaining process stability.

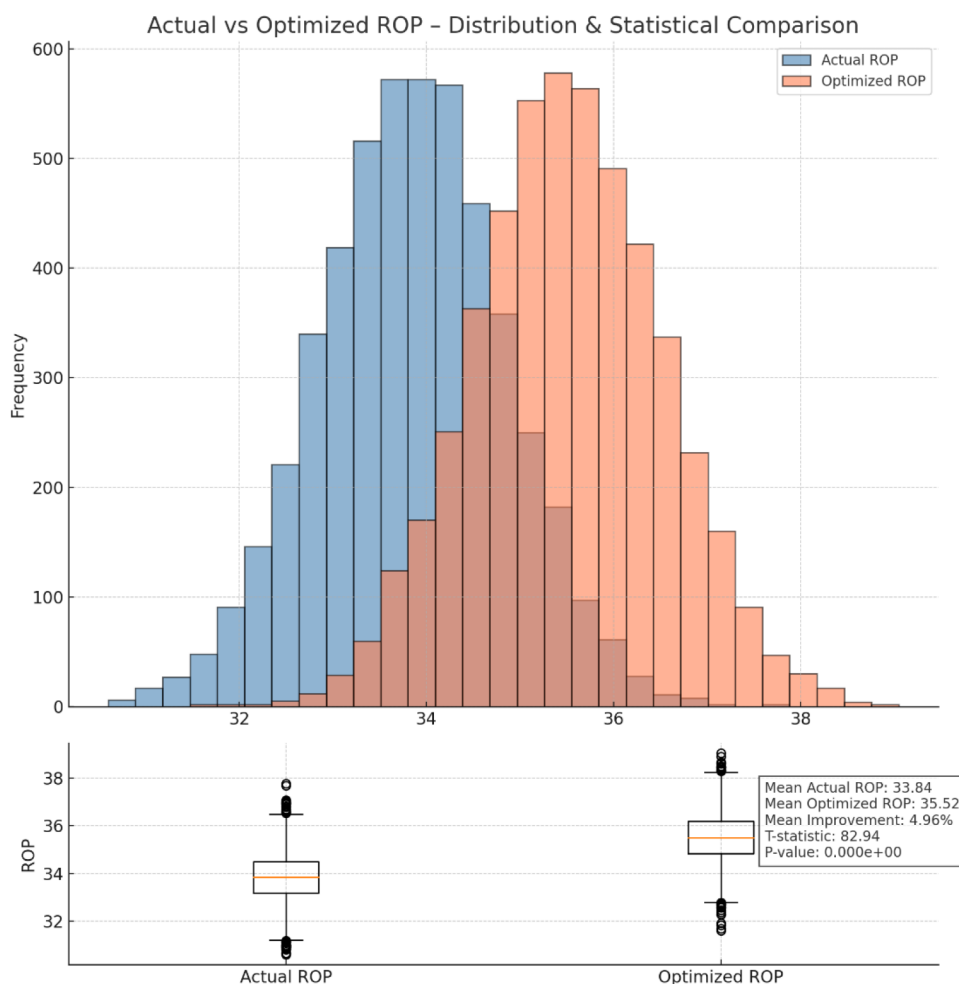


Figure 4—Distribution and Statistic Comparison

The actual versus optimized ROP overlapping distributions show an upward shift in ROP after optimization. A boxplot visualization of the t-test comparison reveals a significant increase in the median ROP for the optimized case, with minimal overlap in the interquartile ranges. This visual representation reinforces the statistically significant results, indicating a substantial improvement in ROP after optimization.

Conclusion and Future Direction

In conclusion, this study addresses the limitations of existing drilling systems, which often struggle to adapt swiftly to evolving downhole conditions. To overcome these challenges, an optimization model was successfully developed and validated to maximize the ROP while maintaining safe FP conditions. By leveraging an open-source drilling dataset and advanced machine learning techniques, predictive models for ROP and FP were built, demonstrating reasonable accuracy as evidenced by low error metrics and strong convergence during training. These models were integrated into a constrained optimization framework, where controllable drilling parameters such as RPM, WOB, and FR were adjusted to maximize ROP under operational and safety constraints. The optimization model significantly improved drilling efficiency, increasing the mean ROP from 33.84 to 35.52, a 5.23% enhancement. This improvement was achieved without introducing significant variability, indicating that the model effectively identifies optimal drilling parameters to enhance performance while maintaining operational stability. The results highlight the potential of the proposed method to provide meaningful insights and improvements in drilling operations.

Future research should focus on expanding the model's applicability by incorporating additional real-time data sources such as downhole telemetry and lithology indicators to improve adaptability under varying formation conditions. Integrating a feedback control mechanism could further enhance the model's responsiveness to unexpected changes during drilling. Exploring hybrid modeling approaches that combine physics-based principles with machine learning may also improve robustness and generalizability across diverse well environments. Finally, validating the model in live field operations would provide critical insights into its performance under real-world constraints and support broader industry adoption.

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